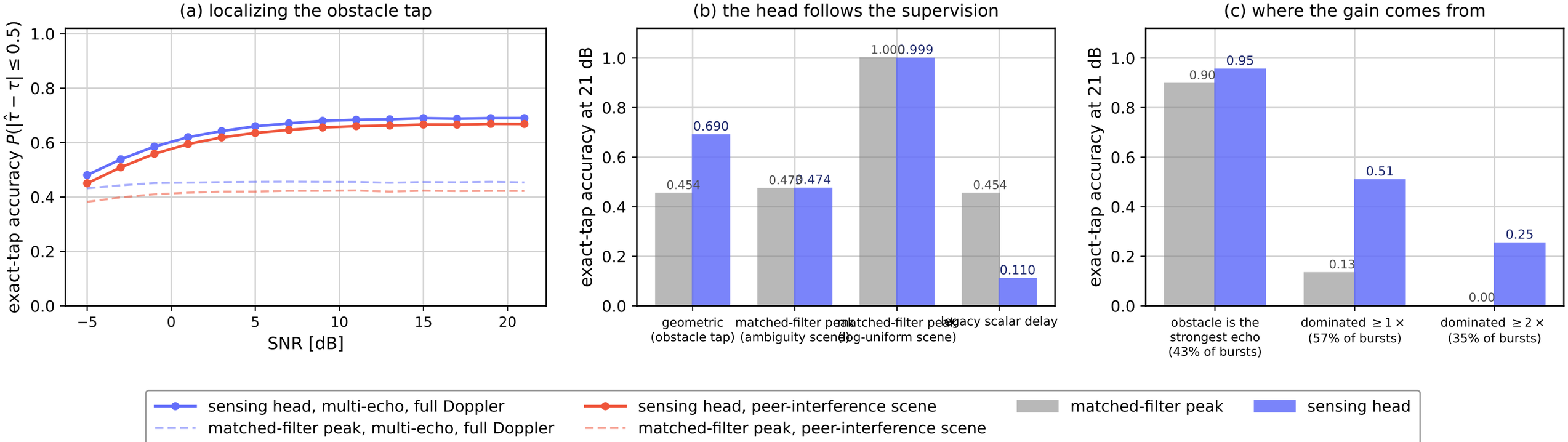
(a) localizing the obstacle tap
exact-tap accuracy $P(|\hat{\tau} - \tau| \le 0.5)$
SNR [dB]
(b) the head follows the supervision
exact-tap accuracy at 21 dB
0.454
0.690
0.478
0.474
1.000
0.999
0.454
0.110
geometric (obstacle tap)
matched-filter peak (ambiguity scene)
matched-filter peak (log-uniform scene)
legacy scalar delay
(c) where the gain comes from
exact-tap accuracy at 21 dB
0.90
0.95
0.13
0.51
0.00
0.25
obstacle is the strongest echo (43% of bursts)
dominated ≥ 1× (57% of bursts)
dominated ≥ 2× (35% of bursts)
sensing head, multi-echo, full Doppler
matched-filter peak, multi-echo, full Doppler
sensing head, peer-interference scene
matched-filter peak, peer-interference scene
matched-filter peak
sensing head